Table 1: **(Q1)** Separation of principal component representations in early (1) vs. late (31) layers (ℓ) of *Llama3* for *Personality* personas. Metrics: Silhouette (Si), Calinski-Harabasz (CH), Euclidean (ED), and Davies-Bouldin (DB). Results are averaged over five seeds (std=0.00, except $\star \approx 0.1$). **Best result** across layers and models. See Appendix Table 5, Table 6, Figure 5, and Table 7 for full results.

Topic	ℓ	SH (↑)	CH (↑)	ED (↑)	DB (↓)
AGREE	1	0.500	340.6*	0.403	0.731
	31	**0.792**	**3264.5**	**27.57**	**0.326**
CONSC	1	0.635	718.8	0.370	0.569
	31	**0.813**	**4150.4**	**27.47**	**0.285**
OPEN	1	0.602	570.2	0.414	0.645
	31	0.795	3564.1	27.60	0.319
EXTRA	1	0.578	527.5	0.382	0.705
	31	**0.788**	**3176.5**	**27.47**	**0.330**
NEURO	1	0.584	615.0	0.378	0.686
	31	**0.796**	**3372.4**	**27.22**	**0.306**

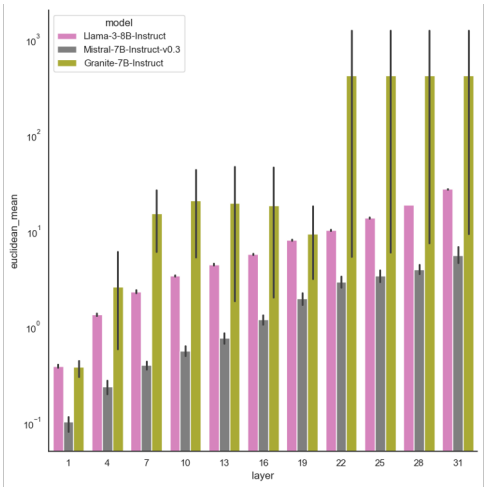


Figure 2: **(Q1)** Euclidean distances between PCA convex hull centroids for MATCHINGBEHAVIOR vs. NOTMATCHINGBEHAVIOR sentences averaged over *Primary Personality Dimensions.*

activations: $S^* = \arg\max_S F(S)$. To efficiently search for this subset Deep Scan uses non-parametric scan statistics (NPSS) [84]. There are three steps to using NPSS on the LLM's activation vectors:

1. **Expectation**: Forming a distribution of "expected" values at each position O_j of the activation vector. We call this expectation our null hypothesis H_0. For instance, we generate the expected distribution over the set of embedding vectors corresponding to NOTMATCHINGBEHAVIOR sentences.
2. **Comparison**: Comparison of embeddings of test set sentences against our expectation H_0. The test set may contain statements from the same distribution as H_0 (e.g., NOTMATCHINGBEHAVIOR) and from the alternative hypothesis H_1 (e.g, MATCHINGBEHAVIOR), which is the hypothesis we are interested in localizing. For each test activation e_{mj}, corresponding to a test sentence X_m and activation position O_j, we compute an empirical p-value. This is defined as the fraction of embeddings from H_0 (Step 1) that exceed the activation value e_{mj}.
3. **Scoring**: We measure the degree of saliency of the resulting test p-values by finding X_S and O_S that maximize the score function F, which estimates how much the observed distribution of p-values from Step 2 deviates from expectation.

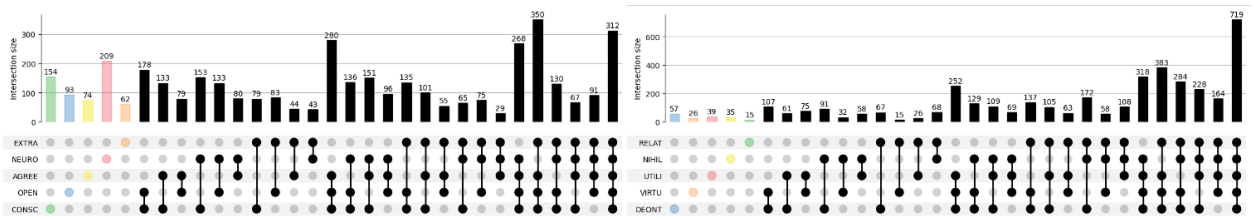
Deep Scan uses an iterative ascent procedure that alternates between: 1) identifying the most persona-driven subset of sentences for a fixed subset of activation units, and 2) identifying the most persona-driven subset of activations that maximizes the score for a fixed subset of sentences. For more details on Deep Scan, refer to prior work [51, 53]. This results in the most persona-driven subset $S^* = X_{S^*} \times O_{S^*}$, where O_{S^*} is the localization of a given persona in our study.

Localization Levels. We localize personas at different levels of granularity, corresponding to different hypotheses H_0 and H_1 (see Table 2): At *Level 2* (inter-persona), we identify activations that differentiate MATCHINGBEHAVIOR from NOTMATCHINGBEHAVIOR sentences within the same persona (e.g., CONS$^+$ vs. CONS$^-$); at *Level 1* (intra-topic), we identify activations distinguishing a specific persona from all other personas within the same topic (e.g., CONS$^+$ vs. {LIBER$^+$ ∪ IMMI$^+$ ∪ LGBTQ$^+$}); at *Level 0* (inter-topic), we identify activations that are common to all personas within a topic and differentiate them from those in other topics (e.g., *Politics*$^+$ vs. {*Ethics*$^+$ ∪ *Personality*$^+$}).

Precision and Recall of Sentences Subset. To validate the usefulness of the identified salient activations O_{S^*}, we report precision and recall of the corresponding subset of sentences identified X_{S^*} with respect to the identification hypothesis H_1. In our context, precision is the fraction of test sentences in X_{S^*} that truly satisfy H_1 (accuracy of our positive detections), and recall is the fraction of test sentences that satisfy H_1 and are included in X_{S^*} (coverage).

5 Results

We now present and discuss our findings related to our research questions, **(Q1)** and **(Q2)**, as outlined in § 3.4. We denote the first layer (simple input layer) as 0, and the last layer as 31.



(a) (b)

Figure 3: **(Q2)** Upset plots illustrating the overlap of sets of salient last-layer activations from MATCHINGBEHAVIOR sentences, as identified by Deep Scan, across personas. Each bar represents the number of activations shared by a specific combination of personas.

(Q1) Which Layers and Models Show the Strongest Signal for Persona Representations?

We first study which layers provide the strongest signals for encoding personas for different LLMs. Specifically, we identify the layer that exhibits the greatest divergence between the principal components (PCs) of the last token representations for sentences corresponding to a given persona—comparing q_+ (MATCHINGBEHAVIOR) and q_- (NOTMATCHINGBEHAVIOR) sentences using the methods described in § 4.1. Our findings lay the groundwork for our next step, where we seek to localize sets of activations within a layer encoding persona information.

Results. Fig. 1b shows the first three PC embeddings for the IMMI persona across several layers, comparing q_+ and q_- embeddings. The PC embeddings overlap considerably in the initial layer, while later layers show increasing separation—with the clearest distinction in the final layer of *Llama3*. We find similar trends for other models and personas (see Appendix Fig. 7 and 8).

We use the metrics described in § 4.1 to quantify the separation between the two embedding groups. Fig. 2 shows the Euclidean distances (for all three models) between the centroids of the convex hulls for the two groups of clusters $q_- \cup q_+$, averaged over *Primary Personality Dimensions* personas. See Appendix Figure 6 for all personas. Across the models, the largest distances are found in the later layers (20–31). Tab. 1 reports additional metrics evaluating the separation, overlap, and compactness of the groups q_- and q_+. Most measures indicate that the final layer of *Llama3* achieves the strongest separation. We find, however, that for some personas, certain metrics favor earlier layers or other models. This suggests that while *Llama3* generally provides the best overall separation, for persona-specific applications, evaluating different metrics and models might be beneficial.

Overall, later layers exhibit the greatest separation between q_+ (MATCHINGBEHAVIOR) and q_- (NOTMATCHINGBEHAVIOR) across LLMs, indicating that persona representations become increasingly refined, with final layers encoding the most discriminative features. This aligns with prior work showing that higher layers capture more contextualized, task-specific information [85]. Among the models tested, *Llama3* demonstrated the strongest separation and most cohesive clusters in its final layer, suggesting it most effectively encodes persona-specific information. Consequently, our subsequent analysis focuses exclusively on the last-layer representations of *Llama3*.

(Q2) Are There Unique Locations of Persona Representations Within Layers?

Next, we investigate whether distinct, consistent activation groups within a layer encode different personas. Building on our previous findings, we compare the last token representations from *Llama3* for MATCHINGBEHAVIOR versus NOTMATCHINGBEHAVIOR sentences. We use Deep Scan (§ 4.2) to identify the activation subsets most indicative of persona-specific information O_{S^*}, which we refer to as *salient activations*.

Results. First, we validate the Deep Scan results as described in § 4.2. In Tab. 2 (*Level 2*), we report precision and recall of the corresponding X_{S^*}. We find high precision and recall for all 14 personas, with the precision ranging from 0.778 (NIHIL) to 0.999 (CONSC) and recall from 0.76 (NEURO) to 0.998 (AGREE). This showcases that the found O_{S^*} contains information needed to detect MATCHINGBEHAVIOR of a sentence for a given dimension.

After successful validation, we examine the overlap of salient activation subsets within personas of the same topic, namely *Ethics* (Fig. 3b), *Politics* (Fig. 1c), and *Personality* (Fig. 3a). Recall that the full embedding vector has a dimension of 4096 activations. For *Ethics* personas, only a small fraction of activations are unique—ranging from 0.37% (15 activations) to 1.39% (57)—indicating that few nodes exclusively differentiate each persona. In contrast, we find a substantial overlap among these personas, with 17.55% (719) of the activation vector shared across all. This suggests strong polysemanticity, where the same activation contributes to multiple ethical representations. In comparison, *Politics*

personas display much lower overlap, with only 9.42% (386) shared activations across all. *Personality* personas show a similarly modest overlap at 7.62% (312). *Politics* personas, however, exhibit a larger set of unique activations per persona, ranging from 2.05% (84) to 5.54% (227). Unique activations for *Personality* personas similarly range from 1.51% (62) to 5.10% (209). These findings suggest that individual *Politics*—and to a slightly lesser extent, *Personality*—personas are characterized by more distinct activation patterns. Overall, we find that some personas reveal clearly unique locations within the last-layer last representations of *Llama3*.

What Are the Activation Interactions Between Groups of Personas?

Now, we shift our focus to understanding whether we can differentiate between groups of specific personas (only using MATCHINGBEHAVIOR sentences) based on their embeddings. Specifically, we are interested if we can: (i) distinguish inter-topics, between personas associated with a particular topic (e.g., *Politics*) from other topics, e.g., $\{$*Ethics* $\cup$ *Personality*$\}$, and ii) distinguish intra-topic between a single persona within a topic (e.g., LIBER) and other personas within the same topic, e.g., $\{$CONS $\cup$ LGBTQ $\cup$ IMMI$\}$. We believe this can provide insights on different levels of granularity that can inform interventions to generate output within a given persona.

Results. We validate our results by reporting the precision and recall of our salient node detection method (see Tab. 2). We achieve high performance at inter-topic *Level 0*. The lowest precision is 0.885 (*Politics*), and the lowest recall is 0.842 (*Ethics*). This suggests that our approach is highly effective at identifying topic-level activation patterns that all personas within a topic share and separating them from personas of other topics.

In contrast, our results are mixed at intra-topic *Level 1*. For 4 of the 14 evaluated personas, we observe high precision (ranging from 0.74 to 0.97) and high recall (ranging from 0.79 to 0.98), indicating reliable detection in these cases. However, for the remaining 9 personas, precision falls (majority ranging from 0.42 to 0.63, with the exception of UTILI at 0.93), and recall is generally lower (ranging from 0.66 to 0.96). This suggests that we can detect broad, inter-topic differences, and patterns are found to be less consistent for intra-topic distinctions—possibly due to overlapping activation patterns or less pronounced differentiating features among some personas.

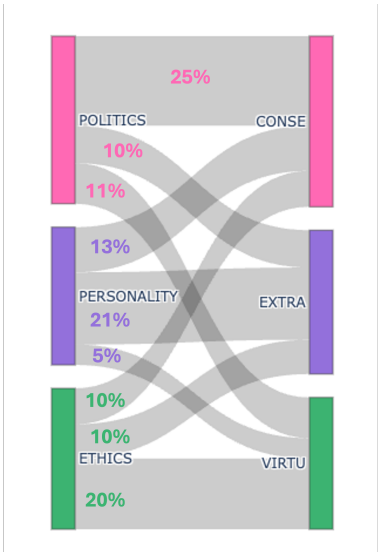


Figure 4: **(Q2)** Overlap of salient activations between topics and sampled persona from each topic.

Given these observations, we focus only on the interplay between salient activations of *Level 0* and *Level 2* in the further analysis. First, at *Level 0*, we find no overlap among salient activations of all three topics—*Ethics*, *Personality*, and *Politics*. In pairwise comparisons, we observe that there is no overlap between *Ethics* and *Personality*, a modest overlap of approximately 7% of activations between *Ethics* and *Politics*, and the largest overlap of roughly 12% between *Politics* and *Personality*. Consequently, the unique nodes attributed to each topic are about 93% for *Ethics*, 88% for *Personality*, and 85% for *Politics*[8]. These findings suggest that distinct activation locations characterize each topic. At the same time, a certain degree of commonality (polysemanticity) remains—particularly between *Politics* and *Personality*—which may reflect shared underlying conceptual features in their representations.

Lastly, we are interested in understanding how the inter-topic activations (*Level 0*) relate to more detailed inter-persona patterns (*Level 2*). In Fig. 4, we show the overlap of salient activations between these levels for one example persona per topic. We observe an overlap of 25% of the salient activations between *Politics* and political persona CONSC. Similarly, for *Personality* and personality trait EXTRA, we find a 21% overlap, and for *Ethics* and ethical persona VIRTUE, a 20% overlap.

These findings suggest that a significant portion of a persona's encoding includes topic information, while the observed overlaps with other persona topics indicate that some activations are shared across these representational spaces.

6 Summary, Limitations, and Future Work

We investigated where LLMs encode persona-related information within their internal representations, analyzing last token activation vectors from 3 families of decoder-only LLMs using persona-specific statements from 14 datasets across *Politics*, *Ethics*, and *Personality* topics. Our PCA showed the strongest signal in separating persona information

[8]For a visualization, see Appendix Fig. 10.